

WASP-36b

R-0809

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_240115 · created 2026-07-28T11:27:26+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460324.9031 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.146 +/- 0.014
Transit depth (Rp/Rs)^2	2.13 +/- 0.4 [%]
Orbital Inclination (inc)	87.99 +/- 1.14
Airmass coefficient 1 (a1)	1.012 +/- 0.012
Airmass coefficient 2 (a2)	-0.008 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.23 %
Best Comparison Star	#3 - [483, 214]
Optimal Aperture	3.51
Optimal Annulus	11.32
Transit Duration (day)	0.0943 +/- 0.0023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.146 ± 0.014 vs 0.1368 (deviation 0.58σ)
Duration fit vs published	0.0943 d vs 0.0758 d (deviation 7.09σ)
Mid-time O-C	-9.5 min
Depth SNR	9.2
Residual scatter	1.23 %

Light curve

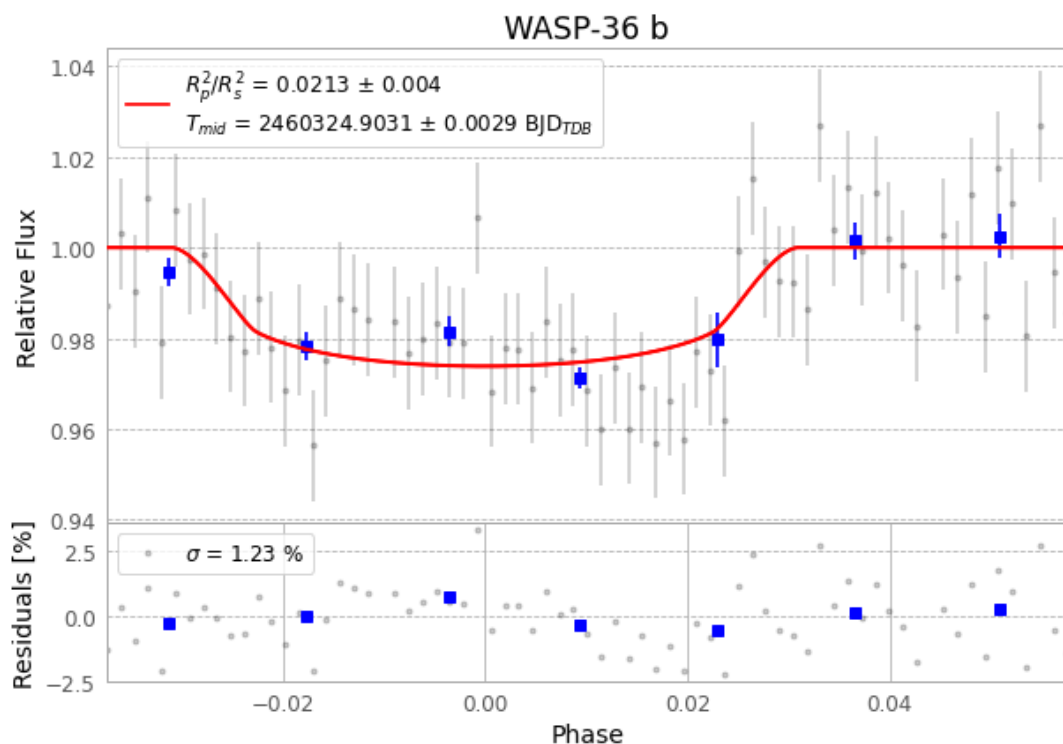


Figure 1 — FinalLightCurve_WASP-36 b_2024-01-15.png (SHA-256 b7ec8c263daf6ea3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [415, 246], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e927d85921f53ced...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

71 FITS frames (47 MB), WASP-36240115080915.FITS → WASP-36240115113915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-240115.log (7412a9d9e8d2...), FinalLightCurve_WASP-36 b_2024-01-15.png (b7ec8c263daf...), FinalLightCurve_WASP-36 b_2024-01-15.csv (28ad4db70d96...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 11:27Z.